

QINGDAO, China

WMO: 548570

Lat: 36.0628N

Lon: 120.3348E

Elev: 77

StdP: 100.40

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-8.8	-6.9	-18.9	0.7	-1.4	-16.9	0.9	-0.9	11.4	-2.6	10.4	-2.2	2.7	340	0.487

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	7.3	33.1	24.3	31.8	23.8	30.2	23.5	27.1	30.0	26.5	29.1	25.9	28.2	4.4	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
26.3	21.9	28.7	25.9	21.4	28.4	25.1	20.4	27.6	87.0	30.0	83.9	29.2	80.9	28.7	30.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
10.2	9.0	7.9	-11.8	35.9	1.9	1.8	-13.1	37.2	-14.2	38.3	-15.2	39.3	-16.6	40.6		
			-12.6	28.5	1.9	0.7	-14.0	29.0	-15.1	29.4	-16.1	29.8	-17.5	30.3		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	13.9	-0.6	2.0	7.2	13.4	19.2	23.1	26.6	22.4	16.3	8.5	1.7	
	DBStd	10.04	3.39	3.71	3.86	3.27	2.89	2.26	2.13	2.16	2.65	3.46	4.21	3.60
	HDD10.0	1001	327	225	103	8	0	0	0	0	2	78	257	
	HDD18.3	2458	586	458	344	151	24	1	0	3	81	296	515	
	CDD10.0	2437	0	1	17	111	285	393	514	513	373	197	32	1
	CDD18.3	851	0	0	0	3	51	144	256	255	126	17	0	0
	CDH23.3	6381	0	0	1	31	322	777	2139	2338	716	57	1	0
	CDH26.7	1902	0	0	0	6	81	233	697	729	151	6	0	0
Wind	WSAvg	3.7	3.5	3.8	4.3	4.5	4.1	3.9	3.5	3.3	3.1	3.3	3.6	3.6
Precipitation	PrecAvg	718	10	14	20	34	56	79	172	153	83	42	29	12
	PrecMax	1371	45	50	93	105	159	252	499	482	402	151	132	37
	PrecMin	305	0	0	0	1	2	3	21	29	2	0	0	0
	PrecStd	251	13	12	18	23	36	57	100	107	78	37	32	11
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	10.2	13.8	21.2	26.9	32.0	34.0	35.2	34.2	31.8	27.1	20.9	13.8
		MCWB	5.4	7.5	11.8	16.4	18.8	21.0	25.4	25.6	21.5	18.2	14.8	10.2
	2%	DB	8.1	11.0	18.0	23.2	28.9	31.1	33.2	32.8	29.9	24.9	18.6	11.8
		MCWB	4.8	5.8	9.9	14.1	17.7	20.4	25.3	25.1	21.2	16.8	13.3	8.5
	5%	DB	6.8	9.1	15.1	21.1	26.8	29.2	31.8	31.2	28.2	23.2	17.1	9.9
		MCWB	3.6	4.7	8.6	13.1	16.8	20.3	24.8	24.8	20.4	16.5	12.9	6.8
	10%	DB	5.2	7.8	13.0	19.0	24.9	27.4	30.1	30.1	27.0	21.9	15.9	8.1
		MCWB	2.4	4.2	7.2	11.8	16.0	19.9	24.6	24.5	20.3	16.3	12.1	5.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	7.5	9.3	13.4	18.1	21.1	25.2	28.3	28.2	25.7	21.0	16.8	11.6
		MCDB	8.6	12.1	18.2	23.4	26.9	29.0	32.3	30.9	27.8	23.3	18.5	12.7
	2%	WB	5.6	7.1	11.2	16.1	19.7	23.7	27.2	27.3	24.3	19.6	15.3	9.0
		MCDB	7.3	9.3	16.0	20.9	25.3	26.9	30.3	29.9	26.7	22.3	17.3	10.9
	5%	WB	4.1	5.8	9.7	14.7	18.6	22.6	26.5	26.7	23.1	18.5	13.9	7.1
		MCDB	6.0	8.0	13.6	18.9	23.7	25.9	29.2	29.2	25.9	21.5	16.4	9.0
	10%	WB	2.7	4.6	8.3	13.4	17.6	21.7	25.9	26.2	22.1	17.5	12.2	5.5
		MCDB	4.9	6.8	11.7	17.1	22.2	24.8	28.3	28.4	25.2	20.7	15.4	7.8
Mean Daily Temperature Range	5% DB	MDBR	9.0	9.5	10.1	10.5	10.5	9.2	7.5	7.3	9.0	9.9	9.5	9.0
		MCDBR	10.4	11.0	12.5	13.1	13.2	11.9	9.5	8.3	10.2	10.6	10.4	10.7
		MCWBR	7.2	6.9	6.8	6.6	5.8	4.7	3.8	3.6	4.5	5.5	6.5	7.7
	5% WB	MCDBR	9.1	9.3	11.3	11.4	11.4	9.1	7.6	6.8	7.7	8.7	8.5	9.1
		MCWBR	7.2	6.7	6.9	6.6	6.0	4.4	3.8	3.4	4.4	5.4	6.5	7.3
Clear-Sky Solar Irradiance	taub	0.503	0.557	0.603	0.617	0.638	0.737	0.721	0.654	0.575	0.599	0.567	0.506	
	taud	1.797	1.713	1.652	1.670	1.672	1.539	1.625	1.744	1.853	1.711	1.730	1.804	
	Ebn at Noon	636	656	674	697	692	625	630	663	691	611	569	598	
	Edh at Noon	167	204	235	243	246	281	256	223	190	201	178	156	
All-Sky Solar Radiation	RadAvg	2.44	3.12	4.45	5.16	5.57	5.19	4.70	4.66	4.25	3.49	2.55	2.19	
	RadStd	0.24	0.36	0.35	0.30	0.42	0.39	0.49	0.53	0.36	0.35	0.28	0.19	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46)	Station Only	N/A	-1.11	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (6 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+66	N/A

Nomenclature: See separate page